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PI Name:	Lelkes, Peter Istvan Ph.D.		
Project Title:	The Effects of Aggregate Culture in Modeled Microgravity and Partial Gravity on Endocrine Differentiation of Induced Pluripotent Stem Cells (iPSC)		
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	The long-term goal of this proposal is to study the effects of modeled microgravity and partial gravity on aggregate (organoid) cultures of induced pluripotent stem cells (iPSCs). Our working hypothesis, based on some preliminary data, is that hypogravity, i.e., modeled microgravity and/or partial gravity, will enhance iPSC differentiation into the endodermal lineages. Specifically, we will compare the efficiency of two distinct microgravity gravity / partial gravity simulating devices, i.e., the Rotating Wall Vessel (RWV) bioreactor and Random Positioning Machine (RPM) to modulate the maintenance of iPSCs stemness and the directed differentiation of iPSC into definitive endoderm. In testing our hypothesis, we developed the following specific aims: Specific Aim 1. To characterize the differentiation of iPSC into definitive endoderm in 2D monolayer culture in normal, and simulated partial- (Moon, Mars) and microgravity; and in 3D aggregate suspension culture in partial- (RPM) and microgravity (RPM and RWV). Specific Aim 2. Examine the underlying genetic, epigenetic, molecular, and biochemical mechanisms of stem cell differentiation and behavior that are affected by aggregate culture, hypoxia, and hypogravity; and identify specific molecular and signaling factors important in modulation of differentiation.
Task Description:	In these studies, we will use biochemical, immunological, and molecular biological approaches to analyze the differential expression of a set of lineage-specific specific marker genes and gene products under altered gravity conditions. Subsequently, we will identify specific genetic, epigenetic, and molecular mechanisms that underlie and are influenced by partial- and microgravity and compare these biologic factors with those from induced iPSC differentiation, in both aggregate and monolayer culture, in standard gravity. The latter data will be analyzed by systems- / bioinformatics-based analytical approaches, such as principle component analysis and network analysis, which will help to identify specific hypogravity-sensitive cellular signaling molecules and pathways that may be differentially affected by varying levels of partial/microgravity, and which could be further explored in actual space experiments. In addition to having available both hypogravity simulating devices, we have introduced several technical innovations, such as modifications in the RWV bioreactors, enhanced oxygenation capabilities in RPM culture, and a high-resolution optical monitoring system that allows real-time visualization of intracellular signaling events. We anticipate that we will demonstrate a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies will contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We anticipate that we will demonstrate a superior environment for iPSC differentiation in simulated and/or partial gravity conditions and that knowledge gained from these studies will contribute to technological and medical advancements both on Earth and in space and enhance our understanding of the adaptation and function of biological processes in altered gravity.
	Originally, we proposed to focus on culturing human induced pluripotent cells (hiPSC) in the modeled microgravity (MMG) environment of the Rotating Wall Vessel (RWV) Bioreactor system and examine the following culture conditions: 1) 2D versus 3D aggregate culture, 2) normal gravity versus MMG, and 3) maintenance of pluripotency and directed differentiation (DD) of hiPSC into definitive endoderm (DE). To assure success for our studies we made several changes in the experimental protocols. As outlined in the Progress Report from May 2020, we initially had difficulties culturing hiPSC and, as a result, concomitantly examined mouse embryonic stem cells (mESC) as a feasibility study. That study is now completed. A peer-reviewed research paper entitled “Enhanced Induction of Definitive Endoderm Differentiation of Mouse Embryonic Stem Cells in Simulated Microgravity” was published in ‘Stem Cells and Development’ in October 2020. The global Covid-19 pandemic and the ensuing bottleneck in the supply chains certainly delayed our progress on many fronts. However, we regained momentum by making several major adjustments (see below), which have greatly enhanced our research, and put us back on track to publish some of the most recent research using hiPSC. • In our hands, the viability and survival of hiPSC in the RWV were poor. Instead, we used the Random Positioning Machine (RPM; Airbus Defense and Space Netherlands) for generating MMG conditions. • Due to the relatively low efficiency of inducing endodermal differentiation in hiPSC with the small molecule activin (which was an effective inducer for mESC), we used the StemDIFF DE media (StemCell). • We switched vendors for the definitive endoderm (DE) flow cytometry antibodies to increase the number of factors we could analyze, increase fluorescence, and reduce nonspecific binding. • Instead of culturing the cells in T-flask, we switched to using VueLife cell culture bags, which proved to be the superior choice, in terms of reducing the necessary culture volume and hence the expenses for the costly stem cell media and additives Experimental Design. To study the effects of modeled microgravity of definitive endoderm differentiation of hiPSC, we examined the following eight experimental conditions: 2D Control (hiPSC in colony culture in pluripotency maintenance media); 2D DE(directed differentiation in 2D culture into DE); 3D Control (EBs in maintenance media); 3D DE (EBs in DE media); 2D Control microG (2D culture in MMG on the RPM); 2D DE microG; 3D Control microG (EBs in the RPM); 3D DE microG. Culture protocols and morphology: 2D and 3D Culture. Human iPSCs (ATCC) were maintained feeder-free in MatrigelTM- coated flasks in mTeSR-Plus maintenance media (StemCell). For 2D culture, cells were plated into either culture flasks (normal gravity) or Matrigel-coated VueLife cell culture begs. Colony morphology was similar in flasks and VueLife bags For 3D organoid culture, we added 7500 cells/well into low-attachment 96-well plates (Corning) and cultured them in maintenance media for 2 days. In this system, the cells aggregated into 3D structures called embryoid bodies (EBs) without attaching to the culture vessel. EBs were gently transferred into either non-tissue culture-treated 60 mm dishes (normal gravity) or non-treated VueLife bags (microgravity). The control cells and EBs were cultured in normal gravity or microgravity for 6 days, as with our feasibility studies with mESCs. Over the culture period, EBs formed consistent, smooth-edged, spherical organoids a few hundred microns in diameter Confocal microscopy showed an outer core of mostly viable cells with Live-Dead staining Differentiation protocol. Cells were directly differentiated to DE using the StemDIFF DE kit (StemCell). As indicated above, our original protocol (which worked for mESC) had

Task Progress:	to be changed for hiPSC due to poor differentiation efficiency with activin A. Cells were plated as single cells with 10 microm ROCK inhibitor Y-27632 and cultured overnight; media formulations were changed daily per manufacturer's protocol. Similar morphologies were seen in flasks and VueLife bags after plating and 6 days of total culture Cells demonstrated dramatically different morphologies under directed differentiation. After 2 days of culture, the directed differentiation protocol began as with cells in 2D. As noted previously with the mESCs, EBs appeared smaller under directed differentiation and were less smooth-edged than those cultured in maintenance media; more detailed histological and morphometrical analysis is currently being performed. The size difference was presumably caused by cell death during the initial phases of differentiation. We saw similar trends with the cells in 2D. Flow cytometry results. Single-cell suspensions were obtained from 2D and 3D cultures by incubation in TrypLE Express (Gibco/ThermoFisher) dissociation enzyme at 37°C. Cells were fixed, permeabilized, and incubated with the following fluorophore-conjugated anti-human antibodies per manufacturer's protocol: three pluripotency (PP) markers PE-Nanog, PerCP-CyTM5.5-Oct3/4, and AlexaFluor647-Sox2 (BD Biosciences); and three endoderm markers PE-FoxA2, APC-Sox17, and Vio Bright FITC-CXCR4 (Miltenyi). Flow cytometry was performed using the BD Accuri C6. At this time, we are completing the 3 analyses of all conditions in triplicate to perform statistical analyses. As expected, following our optimizations, 2D (undifferentiated) cells (2D Control) expressed high levels of the three PP markers, while 2D cells under directed DE differentiation (2D DE) did not. In MMG, cells cultured in maintenance media in 2D (2D Control microg) and 3D (3D Control microg) showed reduced expression of one or more PP markers, suggesting that microgravity induced enhanced the loss of pluripotency in hiPSC. At this time, we are completing more biological repeats to be able to provide statistical analyses. In terms of DE markers, cells in maintenance media (2D Control, 3D Control) expressed low levels, while cells under directed differentiation expressed high levels (2D DE, 3D DE). Cells cultured in maintenance media in 3D under microgravity (3D Control microg) expressed increased levels of DE markers compared to those at normal gravity; suggesting that in addition to loss of pluripotency, hiPSC spontaneously differentiate in microgravity. It is unclear whether they preferentially differentiate towards endoderm. We plan to evaluate alternative differentiation fates (mesoderm, ectoderm) using a broader array of markers. Finally, under directed endodermal differentiation in microgravity (3D DE microg), Sox17 expression decreases. It could be the case that gradients of factors in the differentiation media, e.g., reduced levels in the interior of the EBs, contributes to the decreased efficiency of differentiation. Future genetic, proteomic, and immunofluorescence analyses and should shed light upon these phenomena. Conclusions and Future Approaches. Our data, so far, support our hypothesis that microgravity and 3D culture downregulate pluripotency while upregulating directed endodermal differentiation of hiPSC. For the remainder of our grant, we will a) run more experiments as described to complete the required number of biological replicates to be able to generate statistically meaningful results and b) carry out more complex genetic (RNASeq), proteomic, and immunofluorescence analyses to evaluate alterative fate decisions. At this point, with the implementation of the changes described above, the study is on course and no further changes are anticipated. Currently as suggested in the proposal, we will leverage the ability of the Random Positioning Machine to simulate the gravities found on Moon and Mars for studying the effects of partial gravities on the hiPSC differentiation. Once these investigations are complete, we expect to prepare and submit at least 2 manuscripts for publication, focusing a) on the effect of MMG on DE differentiation of hiPSC and b) comparing the effects of altered gravity (Earth, Moon, MARS, MMG on hiPSC DE differentiation Specific pitfalls. Imaging of cultured single cells and organoids in the VueLife bags with brightfield/phase contrast microscopy is less than optimal due to out-of-focus defects in the bags themselves. This results in poor optical quality of the images. Currently, we are exploring improved imaging techniques such as confocal microscopy and fluorescent staining. EBs were generated as before. Obviously, there was a significant delay in conducting the experiments, as originally planned, primarily due to long-term lab restrictions and significant supply chain disruptions, both due to the ongoing COVID PF6977983 pandemic. Successful optimization of hiPSC and EB culture, the switch to the RPM, and the change in the protocol for directed DE differentiation have resulted in reliable and consistent experiments. Publications and Abstracts Oss-Ronen L, RA Redden, PI Lelkes. Enhanced Induction of Definitive Endoderm Differentiation of Mouse Embryonic Stem Cells in Simulated Microgravity. Stem Cells Dev. 29(19): 1275-1284. 2020. Lelkes PI, L Oss-Ronen, RA Redden. Simulated microgravity enhances the differentiation of cultured mouse embryonic stem cells towards definitive endoderm. TERMIS 6th World Congress, Maastricht, the Netherlands. November 2021 (accepted as oral platform presentation)
Bibliography Type:	Description: (Last Updated: 09/28/2023)
Articles in Peer-reviewed Journals	Oss-Ronen L, Redden RA, Lelkes PI. "Enhanced induction of definitive endoderm differentiation of mouse embryonic stem cells in simulated microgravity." Stem Cells Dev. 2020 Oct 1;29(19):1275-84. https://doi.org/10.1089/scd.2020.0097 ; PMID: 32731794 , Oct-2020